

Ana Lorena Ortiz Loyola

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Data science student (Tecnológico de Monterrey, expected 2027) with hands-on experience building and testing Python-based automation and ML tooling at Deutsche Bundesbank (AnaCredit), plus project experience with machine learning and version-controlled pipelines. Comfortable working across topics such as AI/LLMs, computer vision, and Git-based tooling, and motivated to implement innovative solutions for industrial communication and hazard-detection systems.

Core Competencies

- **AI, LLMs & Machine Learning:** K-Means clustering, Random Forest, SVM, Neural Networks (Keras, TensorFlow, Scikit-Learn); training and evaluating models end-to-end
- **Programming:** Python (Pandas, NumPy, Scikit-Learn), SQL, secondary Java/C++; building and testing automation and data pipelines
- **Version Control & MLOps:** Git-based version control, automated validation of pipeline outputs before roll-out
- **Data Quality & Testing:** Automated deviation and anomaly detection, rule-based validation, systematic testing of pipeline outputs
- **Cloud & AI Tools:** AWS, Azure Cognitive Services; quick to ramp up on new infrastructure and unfamiliar systems
- **Agentic AI & Documentation:** Experience using Claude Code as an AI coding agent to accelerate development; structured technical documentation, Git

Professional Experience

- **Deutsche Bundesbank (AnaCredit)** **Frankfurt am Main, Germany**
Data Engineer / Intern 04/2026–07/2026
 - Built Python/Pandas-based data pipelines with automated validation tests (row counts, referential integrity, aggregate consistency) to verify pipeline outputs before roll-out
 - Implemented automated error detection identifying deviations (duplicate IDs, value-range violations) in reporting data
 - Documented pipeline logic and validation processes clearly for both technical and non-technical stakeholders
 - Significantly reduced response time to data quality issues through end-to-end ownership, from detection to resolution

Projects

- **Tecnológico de Monterrey**
Predictive Modeling – Healthcare Optimization 10/2024–12/2024
 - Analyzed 25+ years of Mexican national health data to identify patient risk profiles and resource bottlenecks
 - Built predictive models (Random Forest, Neural Networks) and K-Means clustering to segment high-risk groups, directly analogous to clustering structured records into meaningful categories

- **Personal Project (GitHub)**
Banking Data Pipeline – Data Cleaning & Audit Log 05/2026
 - Built an end-to-end pipeline for simulated bank transactions (150 accounts) with a normalized SQLite database and foreign-key integrity checks
 - Staged data cleaning with a fully auditable log; automated balance maintenance via SQL triggers; built using **Claude Code** as an AI coding assistant to accelerate implementation
- **Tecnológico de Monterrey**
Lattice-Based Cryptography Research 10/2024–02/2025
 - Implemented and analyzed the complexity of lattice-based cryptographic algorithms (LWE, NTRU) in Python
 - Contributed to a scientific publication analyzing security parameters and runtime behavior across lattice dimensions

Education

- **Tecnológico de Monterrey** **Mexico**
BSc in Data Science 2022–2027
Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship). Topics: data analysis & statistics, machine learning & AI, cryptography & security, NLP, numerical optimization.
- **Universität Göttingen** **Germany**
Exchange Semester 2025–2026
- **Tecnológico de Monterrey** **Mexico**
Abitur 2019–2022
Grade: 1.0. National Physics Olympiad 2020; Honorable Mention, Physics Olympiad 2021.

Languages

- Spanish (native), English (C1, TOEFL iBT 105/120), German (C1).

Publications

- Contributed to a scientific publication on lattice-based cryptography (LWE, NTRU security parameter and runtime analysis), with Prof. A. F. De Abiega L'Eglise, Tecnológico de Monterrey (citation pending publication).

Honors and Awards

- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.
- **National Physics Olympiad** – Mexico (2020).
- **Honorable Mention, Physics Olympiad** (2021).

References

- Available upon request.